

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	M.JAYA KRISHNA
Register Number	192524500

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I **M. JAYA KRISHNA (192524500)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M.JAYAKRISHNA*

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Aim:

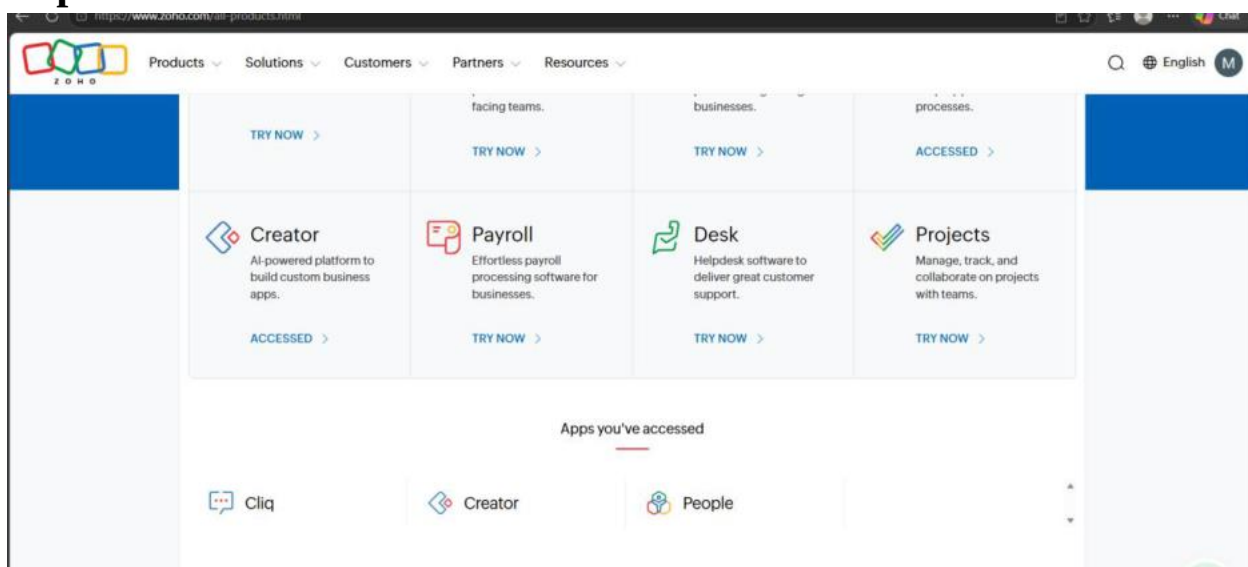
To create a simple **cloud-based Library Book Reservation System** for **SIMATS Library** using **Zoho Creator** as a Software as a Service application.

Steps:

1. Create a new application in Zoho Creator.
2. Create a Library Book Reservation form.
3. Add fields: Title of Book, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.
4. Create a report to view, search, and manage book reservation details.
5. Enter sample book records and verify the reservation system.
6. Test the application using the Zoho Creator cloud platform.

Implemented:

Step1: Goto [zoho.com](https://www.zoho.com) and click the creator



Step 2: Create a new application from scratch in Zoho Creator.

Create New Application



Create using Zia
Build a functional application effortlessly using GenAI

Create



Create from scratch
Create a new application for your business needs

Create



Create from gallery
Pick a pre-built application that suits your requirement

Pick



Import from file
Import your existing data to create an application

Import

Step 3 : Enter Library Book Reservation System as the application name and click Create Application.



Create from scratch

Application Name *

Library Book Reservation System

☐ Enable Environment [Learn More](#)

Create Application

Step 4: Add fields: Title of Book, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.

Library Book Reservation

Trial expires in 7 daysUpgradeEdit this applicationHelp

Library Book ReservationMeghanaM

DashboardReservationsBooksCopy Inventory

Reservations

Book

User

Reservation Date

dd-MM-yyyy

Due Date

dd-MM-yyyy

Status

-Select-

Return Date

dd-MM-yyyy

Created Time

dd-MM-yyyy HH:mm:ss

Last Modified Time

dd-MM-yyyy HH:mm:ss

Submit

Reset

Library Book Reservation

Trial expires in 7 daysUpgradeEdit this applicationHelp

Library Book ReservationMeghanaM

DashboardReservationsBooksCopy Inventory

Books

Title

Author

Publication

Year

#####

Edition

Total Copies

#####

Rack Number

Status

-Select-

Created Time

dd-MM-yyyy HH:mm:ss

Last Modified Time

dd-MM-yyyy HH:mm:ss

Submit

Reset

Steps 5: Create a report to view, search, and manage book reservation details. Enter sample book records and verify the reservation system.

Library Book Reservation

Trial expires in 7 days Upgrade Edit this application Help

Library Book Reservation

Meghana

DashboardReservationsBooksCopy Inventory

Reservations Report

Book

User

Reservation Date

Due Date

Status

Return Date

Created Time

Last Modified Time

1	eve345	19-Jul-2026	26-Jul-2026	Confirmed	26-Jul-2026	19-Jul-2026 08:50:00	19-Jul-2026 08:50:00
2	diana012	18-Jul-2026	25-Jul-2026	Completed	25-Jul-2026	18-Jul-2026 14:20:00	18-Jul-2026 14:20:00
1	charlie789	17-Jul-2026	24-Jul-2026	Expired	24-Jul-2026	17-Jul-2026 11:45:00	17-Jul-2026 11:45:00
1	bob456	16-Jul-2026	23-Jul-2026	Completed	23-Jul-2026	16-Jul-2026 10:15:00	16-Jul-2026 10:15:00
1	alice123	15-Jul-2026	22-Jul-2026	Confirmed	22-Jul-2026	15-Jul-2026 09:30:00	15-Jul-2026 09:30:00

Showing 5 of 5

Library Book Reservation

Trial expires in 7 days Upgrade Edit this application Help

Library Book Reservation

Meghana

DashboardReservationsBooksCopy Inventory

Books Report

Book ID

Title

Author

Publication

Year

Edition

Total Copies

Rack Number

Status

Created Time

Last Modified Time

5	Art of Living	Sarah Green	Art Press	2021	Fifth	45	E-520	Reserved	25-Mar-2026 11:55:00	31-Jul-2026 13:45:00
4	History of Nations	Tom White	History House	2022	Fourth	60	D-415	Checked Out	10-Apr-2026 16:10:00	31-Jul-2026 14:20:00
3	Mystery in the City	Alice Brown	Mystery World	2023	Third	75	C-310	Available	01-May-2026 08:45:00	31-Jul-2026 09:30:00
2	Science of Tomorrow	Bob Smith	Tech Books Inc	2024	Second	100	B-205	Available	20-Jun-2026 14:20:00	31-Jul-2026 11:00:00
1	The Great Adventure	Jane Doe	Fantasy Press	2025	First	50	A-101	Reserved	15-Jul-2026 09:30:00	31-Jul-2026 10:15:00

Library Book Reservation

Trial expires in 7 days Upgrade Edit this application Help

Library Book Reservation

Meghana

DashboardReservationsBooksCopy Inventory

Copy Inventory Report

Book

Copy Number

Status

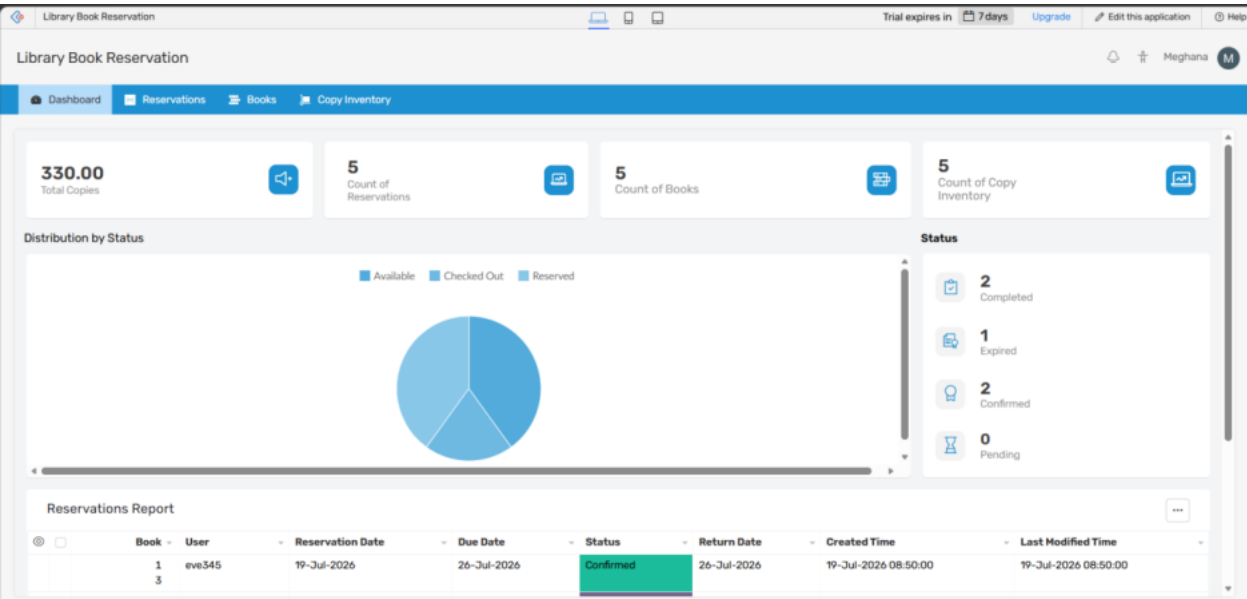
Rack Number

Created Time

Last Modified Time

1	1005	Available	C-01	01-Aug-2026 07:40:00	01-Aug-2026 12:15:00
2	1004	Checked Out	B-02	31-Jul-2026 13:25:00	31-Jul-2026 15:50:00
3	1003	Available	B-01	30-Jul-2026 08:00:00	31-Jul-2026 09:10:00
4	1002	Reserved	A-02	29-Jul-2026 10:20:00	30-Jul-2026 11:45:00
5	1001	Reserved	A-01	28-Jul-2026 09:15:00	28-Jul-2026 14:30:00

Result: The Software has Been Created



The cloud-based Library Book Reservation System was successfully created using Zoho Creator. The system allows users to store, view, and manage library book details and track the Taken/Returned status of books through a Software as a Service-based cloud application.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Aim:

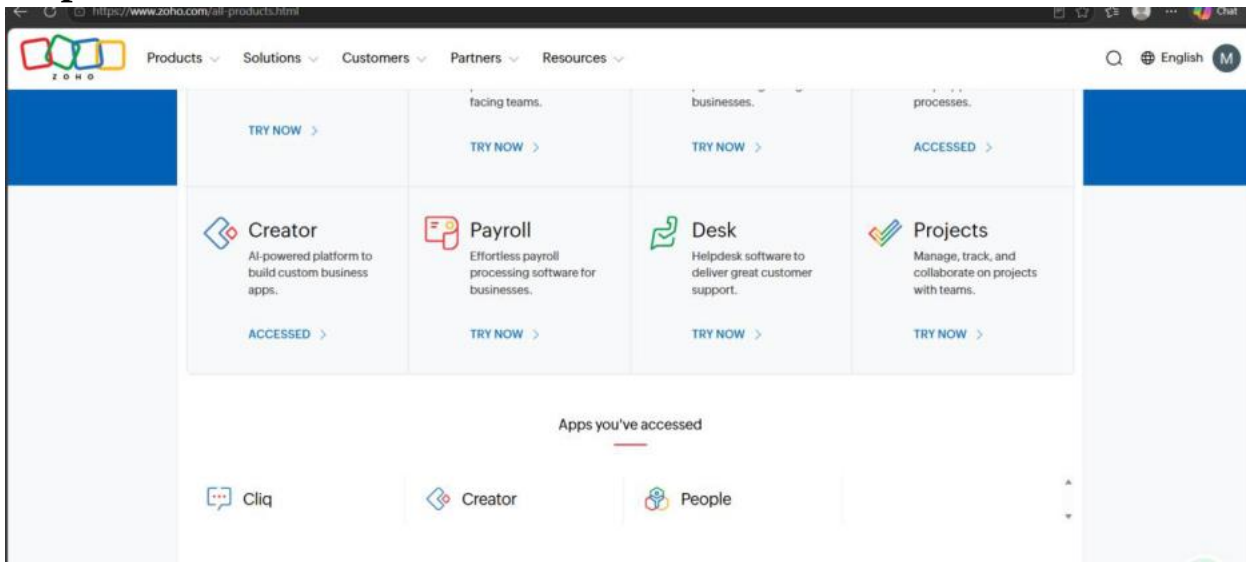
To create a simple cloud-based Payroll Processing System using Zoho Creator to demonstrate Software as a Service (SaaS) and manage employee payroll details.

Steps:

1. Create a new application in Zoho Creator.
2. Create an Employee Payroll form.
3. Add fields: Employee Name, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
4. Create a Payroll Report to view and manage employee salary records.
5. Enter sample employee details and verify the payroll information.
6. Test the application on the Zoho Creator cloud platform.

Implemented:

Step 1: Goto zoho.com and click the creator



Step 2: Create a new application from scratch in Zoho Creator.

Create New Application

 Create using Zia
Build a functional application effortlessly using GenAI [Create](#)


Create from scratch
Create a new application for your business needs
[Create](#)


Create from gallery
Pick a pre-built application that suits your requirement
[Pick](#)


Import from file
Import your existing data to create an application
[Import](#)

Step 3: Enter Payroll Processing System as the application name and click Create Application.




Create from scratch

Application Name *

Payroll Processing System|

☐ Enable Environment [Learn More](#)

Create Application

Step 4: Employee Name, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.

Payroll Processing System

Dashboard

Employee Pay Str...

+

Employee Pay Stru...

Employee Pay Stru...

Payroll Records

Employees

Pay Components

Experience Records

Employee Pay Structure

Employee

Component

Amount

☐

Is Active

Effective Date

Submit

Reset

Payroll Processing System

Trial expires in 7 days Upgrade Edit this application Help

Dashboard

Employee Pay Str...

Payroll Records

+

Payroll Records

Payroll Records Re...

Board

Employees

Pay Components

Experience Records

Payroll Configurat...

Users

Payroll Records

Employee

Pay Period Start

Pay Period End

Basic Pay

HRA

DA

TA

Gross Pay

Net Pay

Processed Date

Status

Submit

Reset

Payroll Processing System

Trial expires in 7 days Upgrade Edit

Dashboard

Employee Pay Str...

Payroll Records

Employees

Pay Components

Experience Records

+

Experience Records

Experience Record...

Payroll Configurat...

Users

Experience Records

Employee

Organization Name

Role

Start Date

End Date

☐

Is Current

Total Experience

Current Org Experience

Submit

Reset

Step 5: Create a Payroll Report to view and manage employee salary records. Enter sample employee details and verify the payroll information

Payroll Processing System

Employee Pay Structure Report

Employee	Component	Amount	Is Active	Effective Date
1	1	\$ 2,750.25	true	10-Aug-2026
2	1	\$ 4,000.00	true	31-Jul-2026
2	2	\$ 1,850.00	false	01-Jan-2025
1	1	\$ 3,200.50	true	15-Jul-2026
2	1	\$ 2,500.00	true	01-Jul-2026
3	2			
3	3			

Showing 5 of 5

Show Summary

Payroll Processing System

Payroll Records Report

Employee	Pay Period Start	Pay Period End	Basic Pay	HRA	DA	TA	Gross Pay	Net Pay	Processed Date	Status
1	01-Jul-2026	31-Jul-2026	\$ 41,000.00	\$ 12,300.00	\$ 2,800.00	\$ 2,200.00	\$ 58,300.00	\$ 49,100.00	31-Jul-2026	Revised
2										
4										
1	01-Jul-2026	31-Jul-2026	\$ 61,000.00	\$ 18,300.00	\$ 4,000.00	\$ 3,500.00	\$ 86,800.00	\$ 72,900.00	31-Jul-2026	Processed
2										
3										
1	01-Jul-2026	31-Jul-2026	\$ 38,000.00	\$ 11,400.00	\$ 2,600.00	\$ 2,000.00	\$ 54,000.00	\$ 45,800.00	31-Jul-2026	Processed
2										
3										
1	01-Jul-2026	31-Jul-2026	\$ 52,000.00	\$ 14,400.00	\$ 3,500.00	\$ 3,000.00	\$ 73,900.00	\$ 61,500.00	31-Jul-2026	Cancelled
2										
3										
1	01-Jul-2026	31-Jul-2026	\$ 45,000.00	\$ 12,000.00	\$ 3,000.00	\$ 2,500.00	\$ 62,500.00	\$ 51,200.00	31-Jul-2026	Revised
2										
4										

Showing 5 of 5

Show Summary

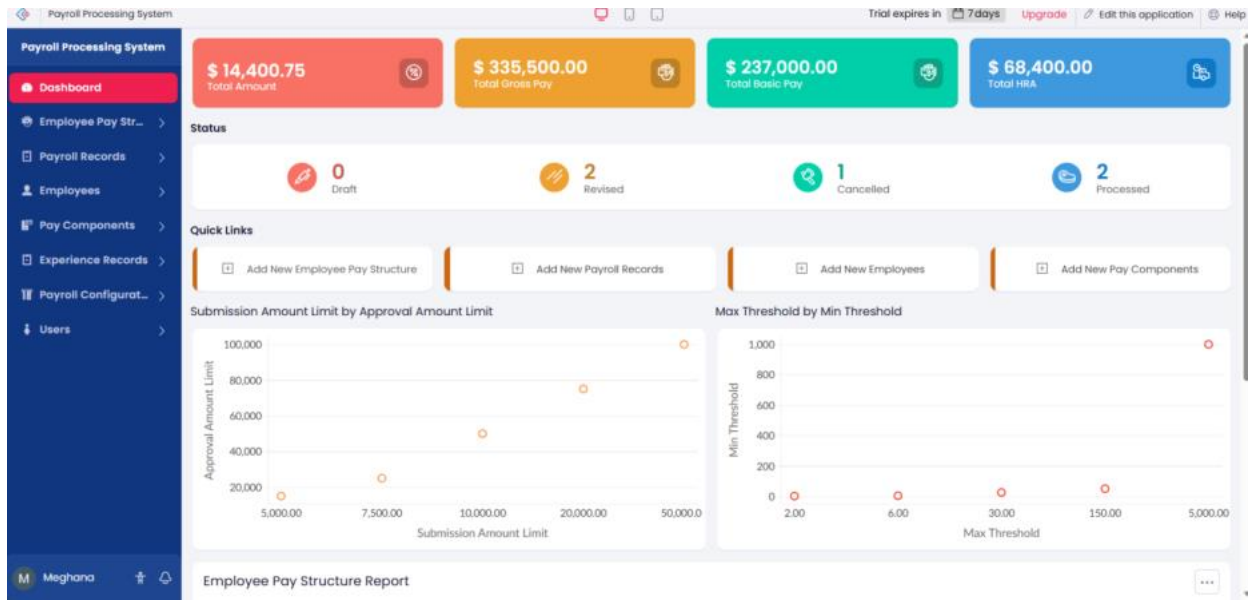
Payroll Processing System

Payroll Configurations Report

Component	Formula	Min Threshold	Max Threshold	Effective Date	Is Active
2	Overtime Multiplier	1.50	2.00	31-Jul-2026	true
1	Retirement Contribution	4.00	6.00	31-Jul-2026	false
2					
3					
1	Health Insurance Deduction	50.00	150.00	31-Jul-2026	true
2					
3					
1	Bonus Calculation	1000.00	5000.00	31-Jul-2026	true
2					
3					
1	Standard Tax Rate	25.50	30.00	31-Jul-2026	true
2					
3					

Showing 5 of 5

Result: The Software has Been Created



The cloud-based Payroll Processing System was successfully created using Zoho Creator. The application allows employee and salary details to be stored, viewed, and managed through a Software as a Service-based cloud platform.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

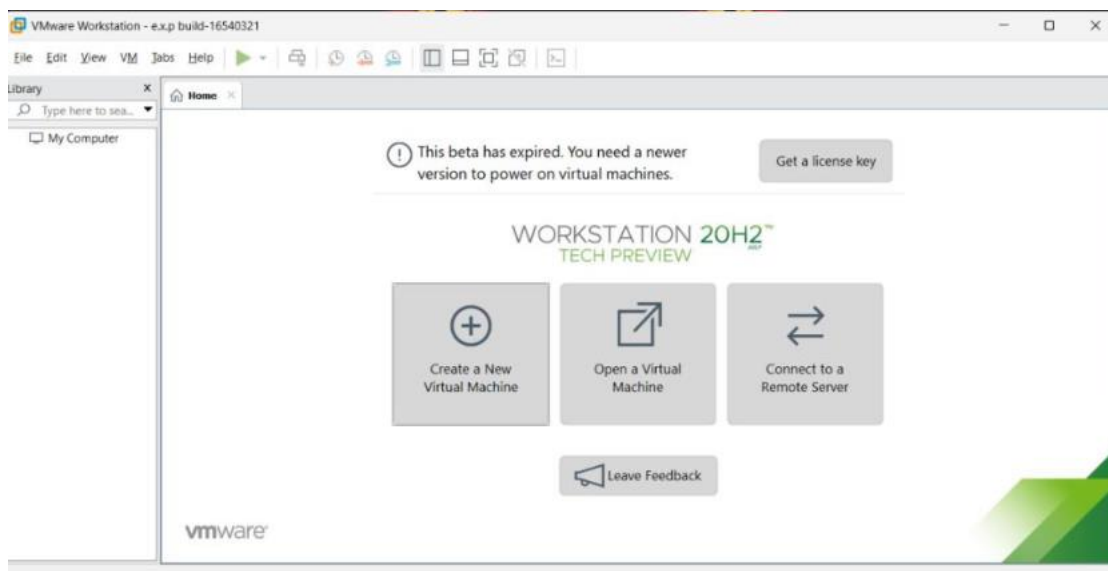
To demonstrate virtualization by installing a Type-2 Hypervisor (VMware Workstation) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage.

Steps:

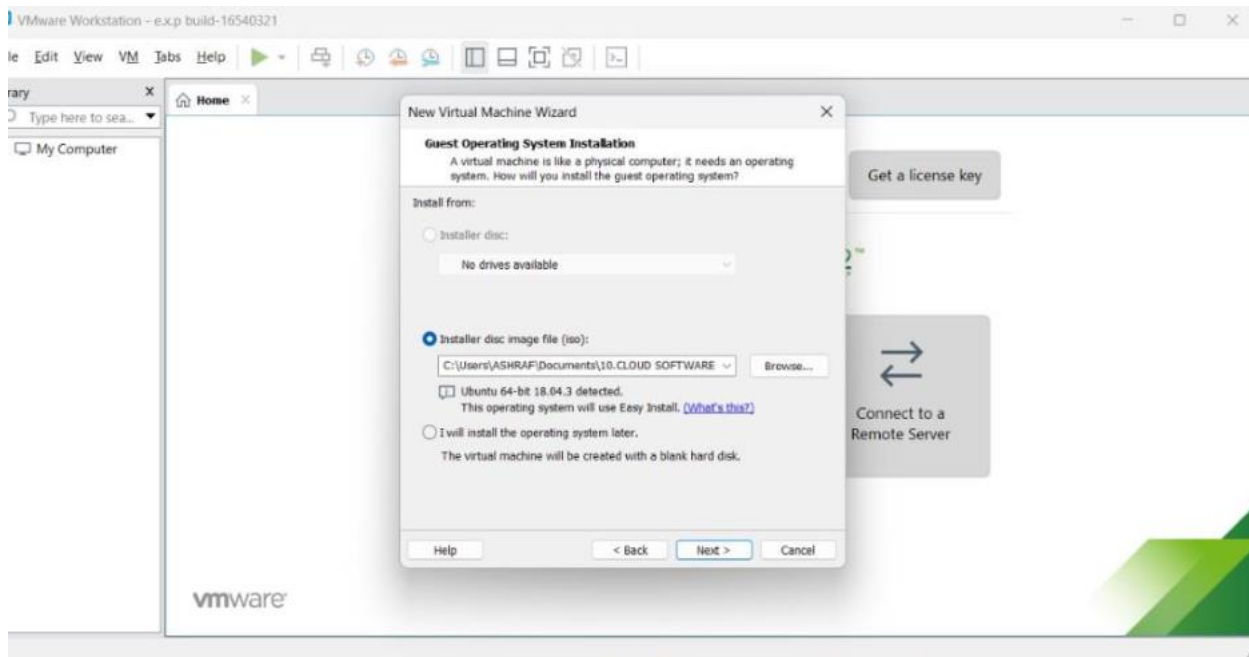
1. Open VMware and select Create New Virtual Machine.
2. Select the Ubuntu ISO file.
3. Set the VM name as Ubuntu-VM.
4. Set RAM = 2 GB and CPU = 1.
5. Set the hard disk size = 15 GB.
6. Start the VM and install Ubuntu.

Implemented:

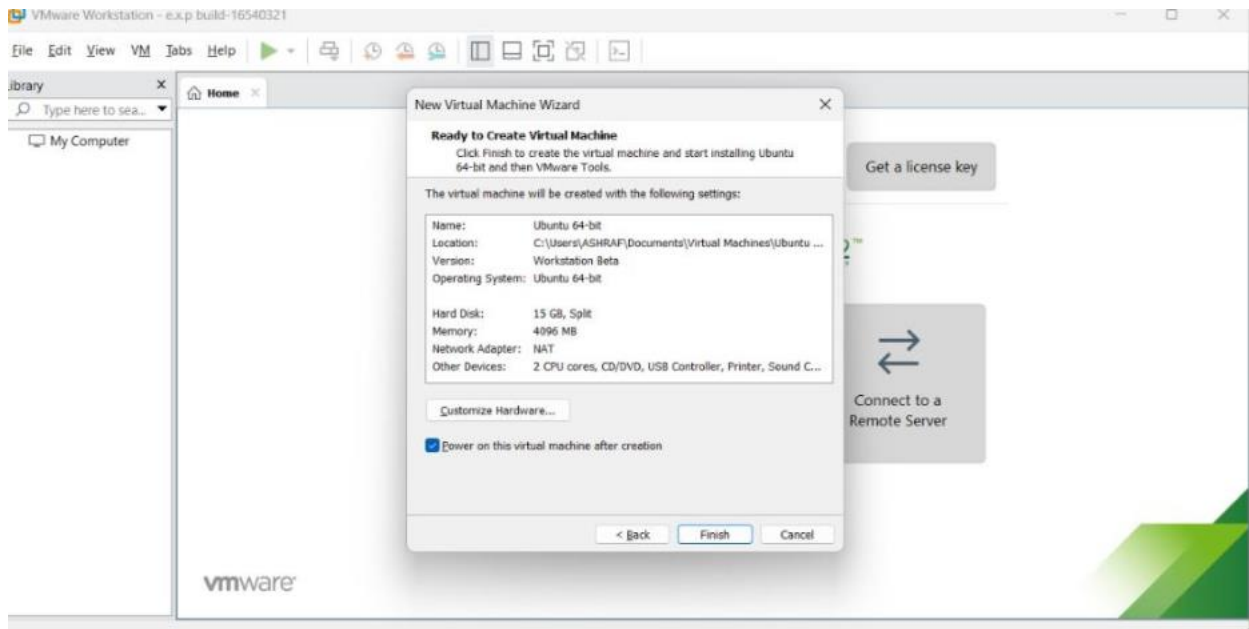
1. Open VMware and select Create New Virtual Machine.



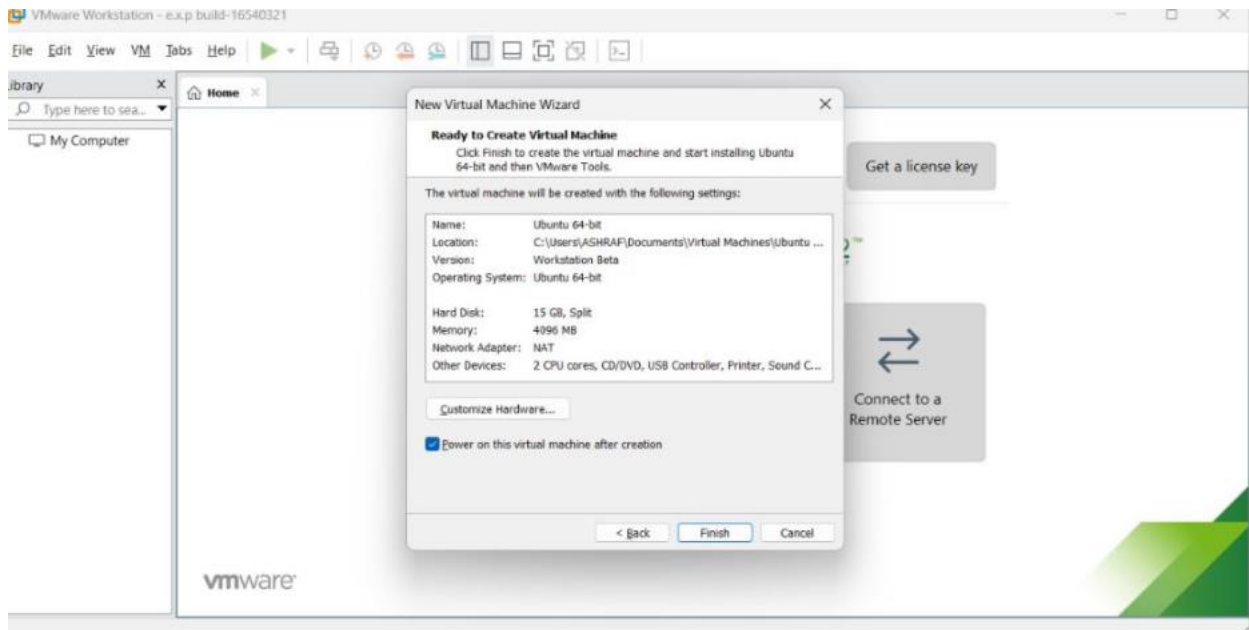
2. Select the Ubuntu ISO file.



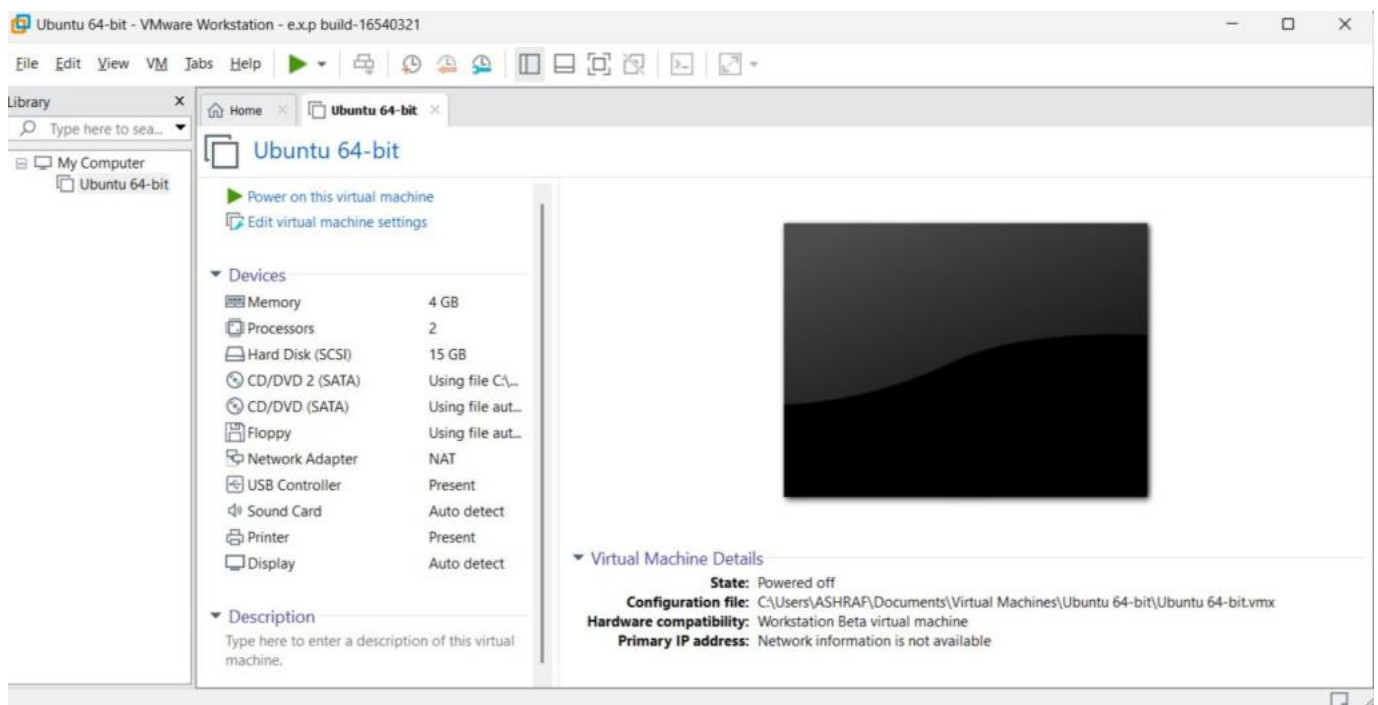
3. Set the VM name as Ubuntu-VM.



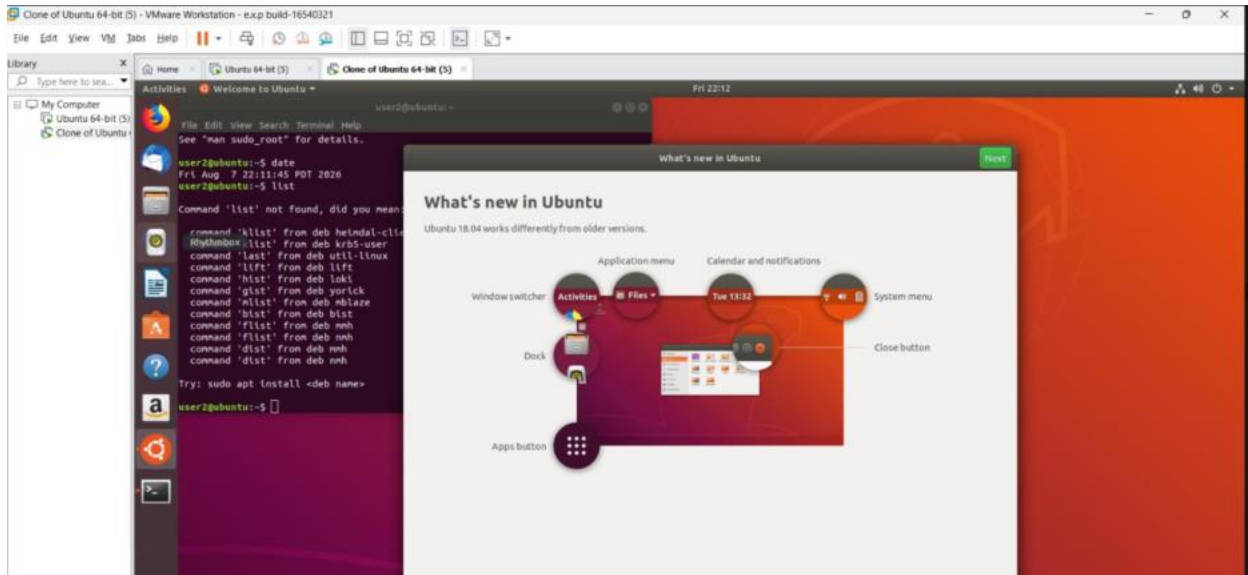
4. Set RAM = 2 GB and CPU = 1.



5. Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

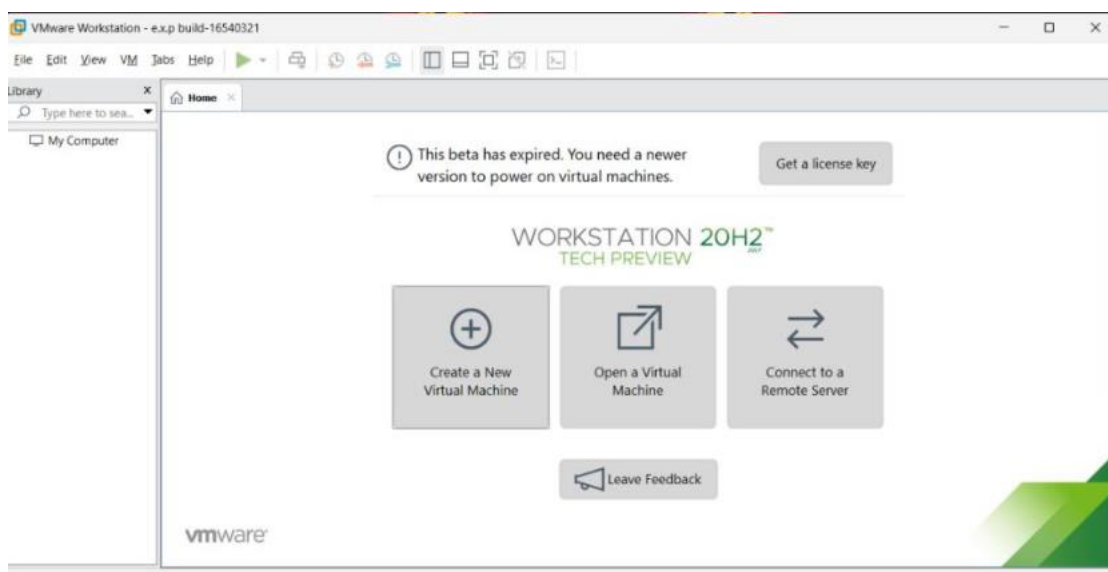
Aim

To demonstrate **virtualization** by installing a **Type-2 Hypervisor (VMware Workstation)**, creating a Virtual Machine with a Windows/Linux operating system, and **cloning the VM** to create and test a previous version of the Virtual Machine.

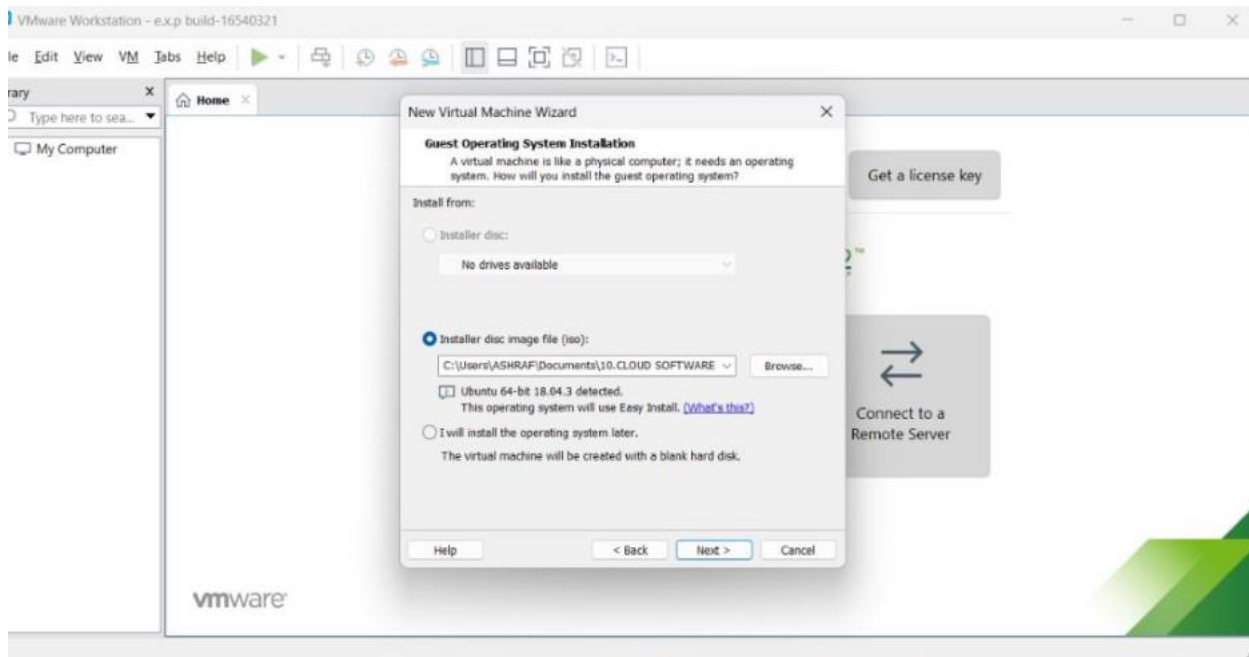
Steps

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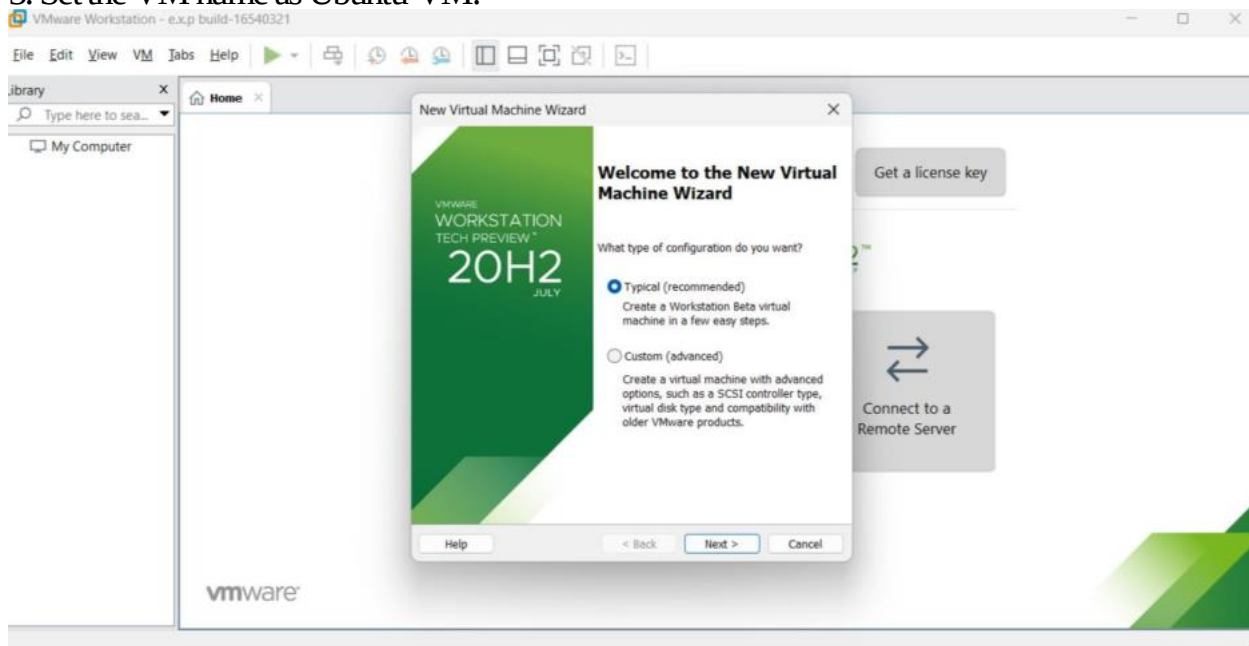
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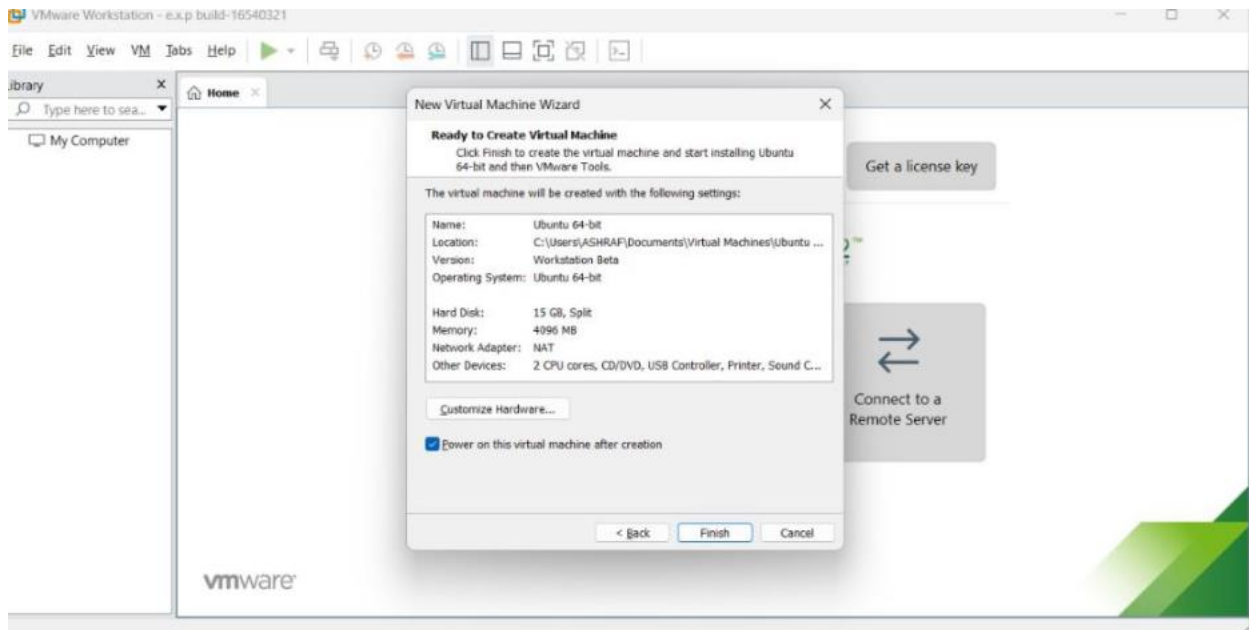
2. Select the Ubuntu ISO file.



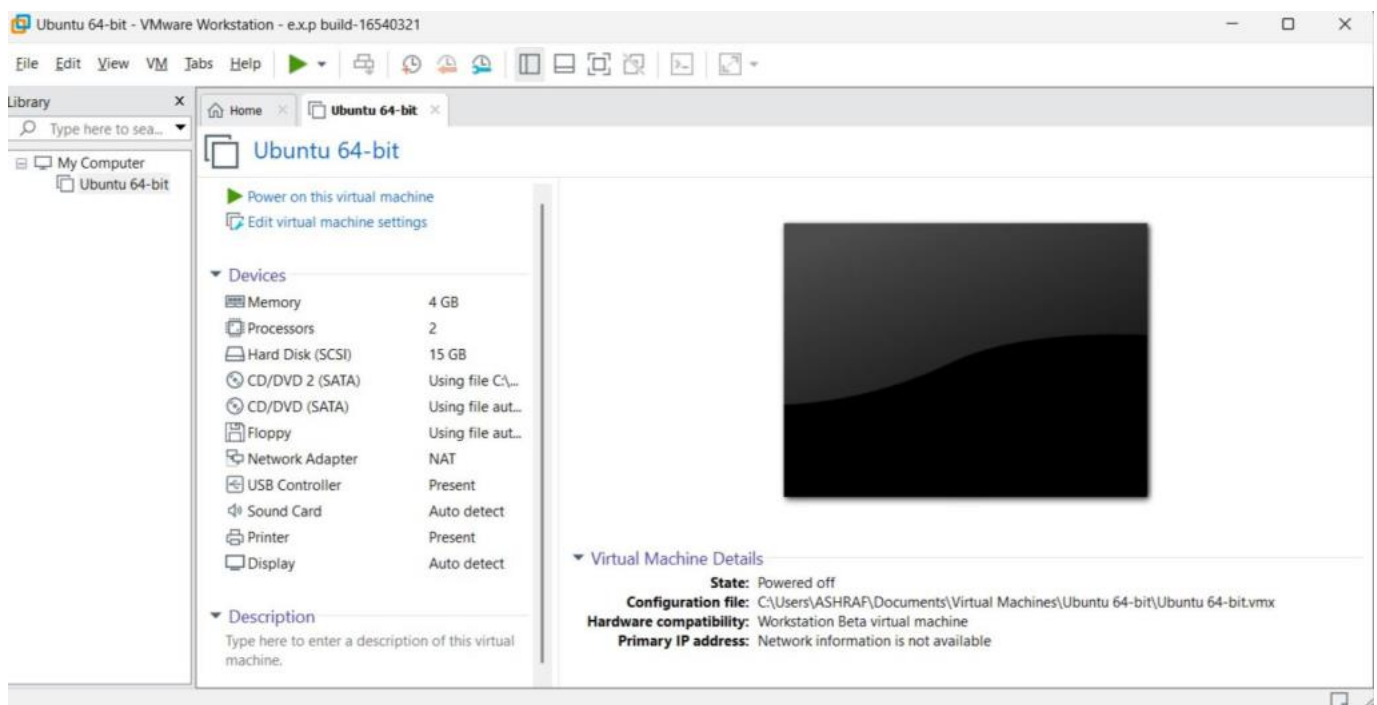
3. Set the VM name as Ubuntu-VM.



4. Set RAM = 2 GB and CPU = 1.



5. Set the hard disk size = 15 GB.



6.Start the VM and install Ubuntu.

